Name: T. MOHANA

Hall Ticket:2303A52446

Batch: 41

Task 1: Initial API Setup for a Service Platform

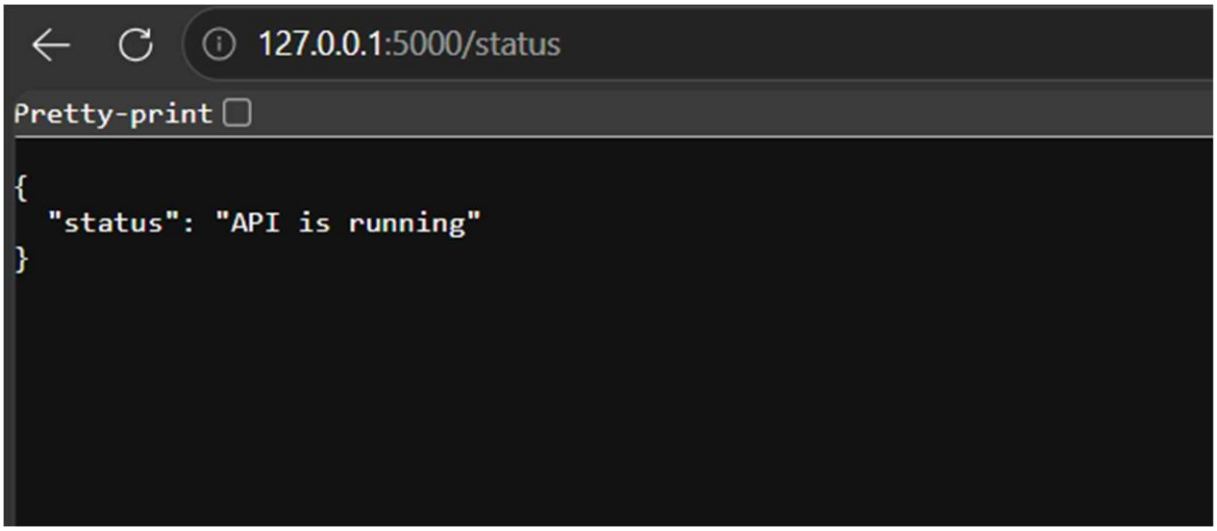
Prompt:

Generate a simple Flask API with a /status endpoint that returns JSON {"status": "API is running"}.
Include code to run the server locally with debug mode enabled.

Code:

```
#Generate a simple Flask API with a /status endpoint that returns JSON {"status": "API is
running"}.Include code to run the server locally with debug mode enabled.

from flask import Flask, jsonify
app = Flask(__name__)

@app.route('/status', methods=['GET'])
def get_status():
    return jsonify({"status": "API is running"})

if __name___ == '__main__':
    app.run(debug=True)
```

Output:



Explanation:

This Flask app creates a basic backend server with one endpoint /status to confirm the API is working.
When accessed, it returns a JSON response indicating the server is running successfully.

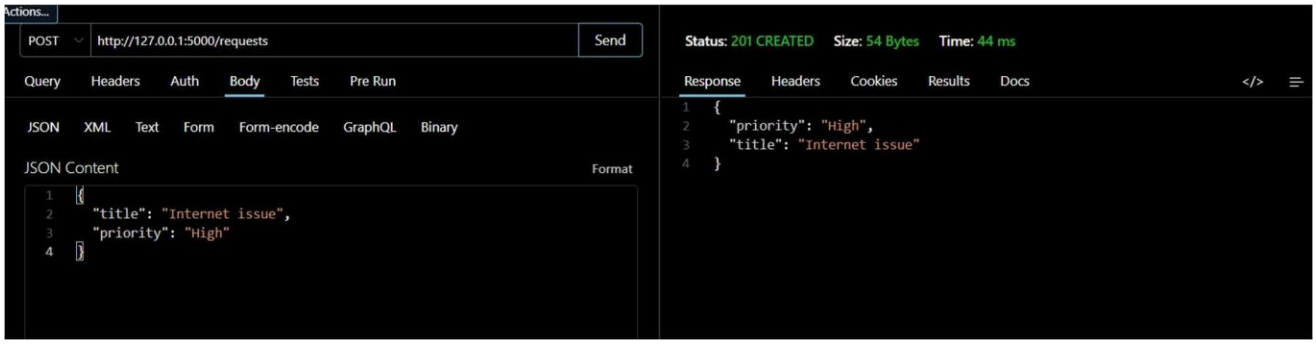
Task 2: **Create API to Add and View Resources**

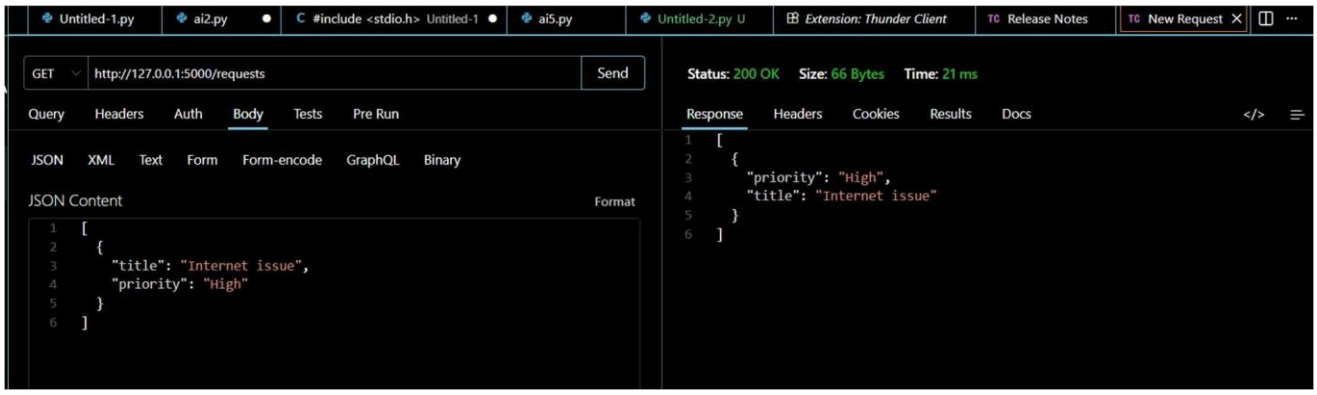
Prompt:

Generate Flask API endpoints: GET /requests to return all requests and POST /requests to add a new request with title and priority.Store data in memory (list) and return JSON responses.

Code:

```
#Generate Flask API endpoints: GET /requests to return all requests and POST /requests to
add a new request with title and priority.Store data in memory (list) and return JSON
responses.
from flask import Flask, request, jsonify
app = Flask(__name__)
requests = []

@app.route('/requests', methods=['GET'])
def get_requests():
    return jsonify(requests)

@app.route('/requests', methods=['POST'])
def add_request():
    data = request.get_json()
    request_item = {
        "title": data.get("title"),
        "priority": data.get("priority")
    }
    requests.append(request_item)
    return jsonify(request_item), 201

if __name___ == '__main__':
    app.run(debug=True)
```





Output:

```
Pretty-p  Refresh (Ctrl+R)
[
  {
    "priority": "High",
    "title": "Internet issue"
  }
]
```

Explanation:

This API stores service requests in a temporary list and allows adding and viewing them using POST and GET endpoints. Data is not permanent restarting the server will erase all stored requests.

Task 3: Modify Existing Resource Using AI Assistance

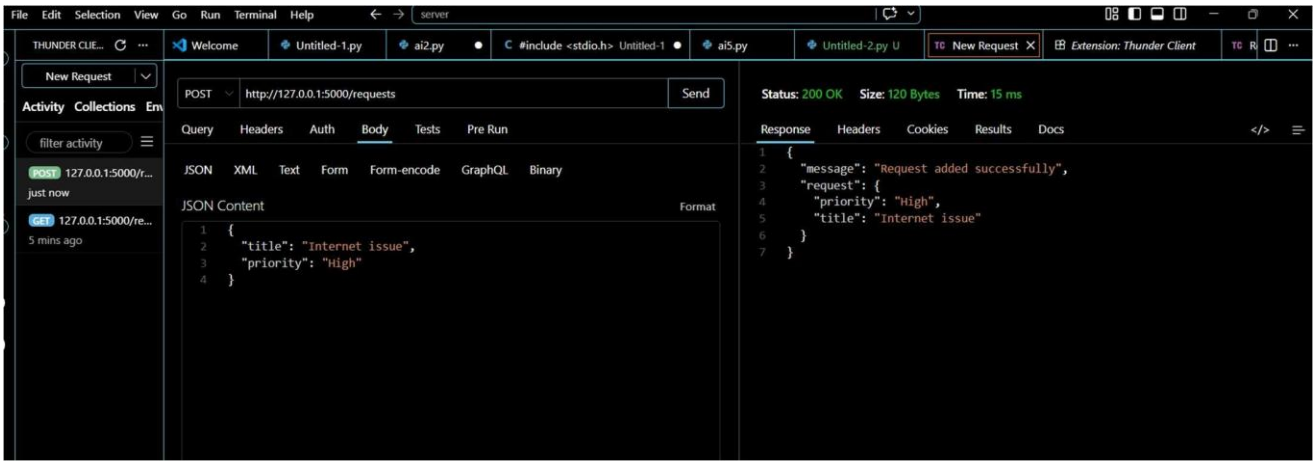
Prompt:

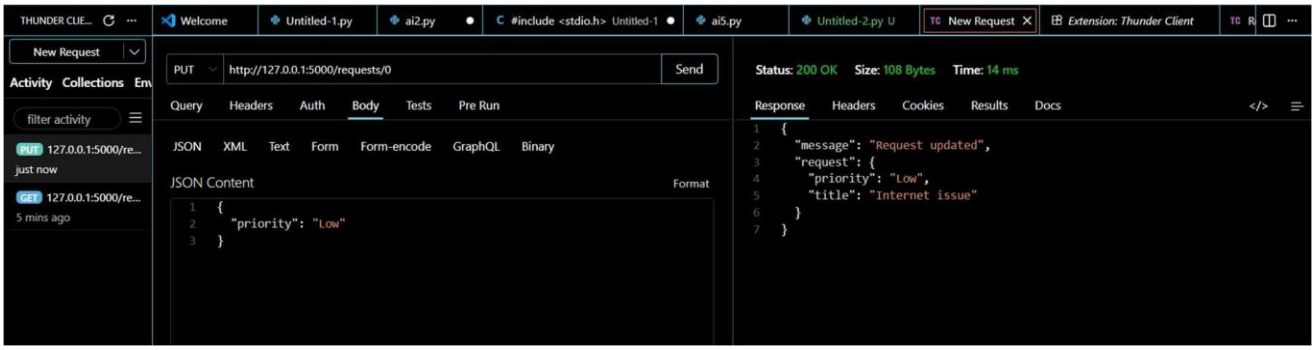
Generate a Flask PUT endpoint /requests/<int:index> to update a request's title and priority by index.
Return updated data on success and a 404 error if the index is invalid.

Code :

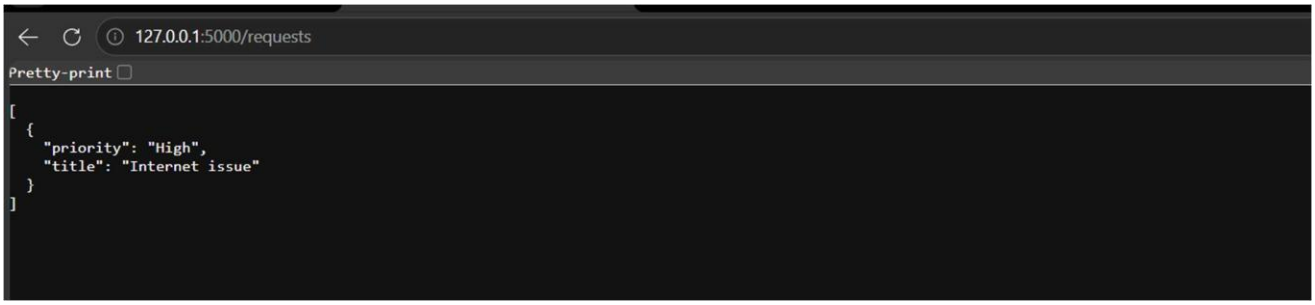
```
#Generate a Flask PUT endpoint /requests/<int:index> to update a request's title and
priority by index.Return updated data on success and a 404 error if the index is invalid.
from flask import Flask, request, jsonify

app = Flask(__name__)

# In-memory storage
requests = []

# STATUS
@app.route('/status')
def status():
    return jsonify({"status": "API is running"})

# GET all requests
@app.route('/requests', methods=['GET'])
def get_requests():
    return jsonify(requests)

# POST new request
@app.route('/requests', methods=['POST'])
def add_request():
    data = request.get_json()

    new_request = {
        "title": data.get("title"),
```

```python
        "priority": data.get("priority")
    }

    requests.append(new_request)

    return jsonify({
        "message": "Request added successfully",
        "request": new_request
    })

# PUT update request
@app.route('/requests/<int:index>', methods=['PUT'])
def update_request(index):
    if index < 0 or index >= len(requests):
        return jsonify({"error": "Request not found"}), 404

    data = request.get_json()

    requests[index]["title"] = data.get("title", requests[index]["title"])
    requests[index]["priority"] = data.get("priority", requests[index]["priority"])

    return jsonify({
        "message": "Request updated",
        "request": requests[index]
    })

if _name___ == '_main_':
    app.run(debug=True)
```





Output:



Explanation:

This endpoint updates an existing request using its index in the list and modifies only provided fields.
It validates the index and returns a 404 error if the request does not exist.

Task 4: Remove a Resource Safely

Prompt:

Generate a Flask DELETE endpoint /requests/<int:index> to remove a request by index and return the deleted object.
Handle invalid indexes safely and return a 404 error with a clear message.

Code:

```
from flask import Flask, request, jsonify

app = Flask(__name__)

# In-memory storage
requests = []

# STATUS
@app.route('/status')
def status():
    return jsonify({"status": "API is running"})

# GET all requests
@app.route('/requests', methods=['GET'])
def get_requests():
    return jsonify(requests)
```

```python
# POST new request
@app.route('/requests', methods=['POST'])
def add_request():
    data = request.get_json()

    new_request = {
        "title": data.get("title"),
        "priority": data.get("priority")
    }

    requests.append(new_request)

    return jsonify({
        "message": "Request added successfully",
        "request": new_request
    })

# PUT update request
@app.route('/requests/<int:index>', methods=['PUT'])
def update_request(index):
    if index < 0 or index >= len(requests):
        return jsonify({"error": "Request not found"}), 404

    data = request.get_json()

    requests[index]["title"] = data.get("title", requests[index]["title"])
    requests[index]["priority"] = data.get("priority", requests[index]["priority"])

    return jsonify({
        "message": "Request updated",
        "request": requests[index]
    })
@app.route('/requests/<int:index>', methods=['DELETE'])
def delete_request(index):
    if index < 0 or index >= len(requests):
        return jsonify({"error": "Request not found"}), 404

    deleted_request = requests.pop(index)

    return jsonify({
        "message": "Request deleted",
        "request": deleted_request
    })

if __name__ == '__main__':
    app.run(debug=True)
```

DELETE http://127.0.0.1:5000/requests/0 Send

Query Headers Auth Body Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

JSON Content Format

```
{
  "title": "Internet issue",
  "priority": "High"
}
```

Output:

C:\Users\lilac\New folder\fashion-store\server\ai5.py

Status: 200 OK Size: 109 Bytes Time: 10 ms

Response Headers Cookies Results Docs

```
{
  "message": "Request deleted",
  "request": {
    "priority": "High",
    "title": "Internet issue"
  }
}
```

Explanation:

This endpoint removes a request from the list using its index and returns the deleted data.
It prevents crashes by validating the index and returning a 404 error if the request does not exist.

Task 5: AI-Assisted API Documentation and Readability

Prompt:

Add detailed docstrings and inline comments to all Flask API endpoints describing method, inputs, and responses clearly. Optionally integrate Swagger documentation using Flask-RESTX to auto-generate API docs UI.

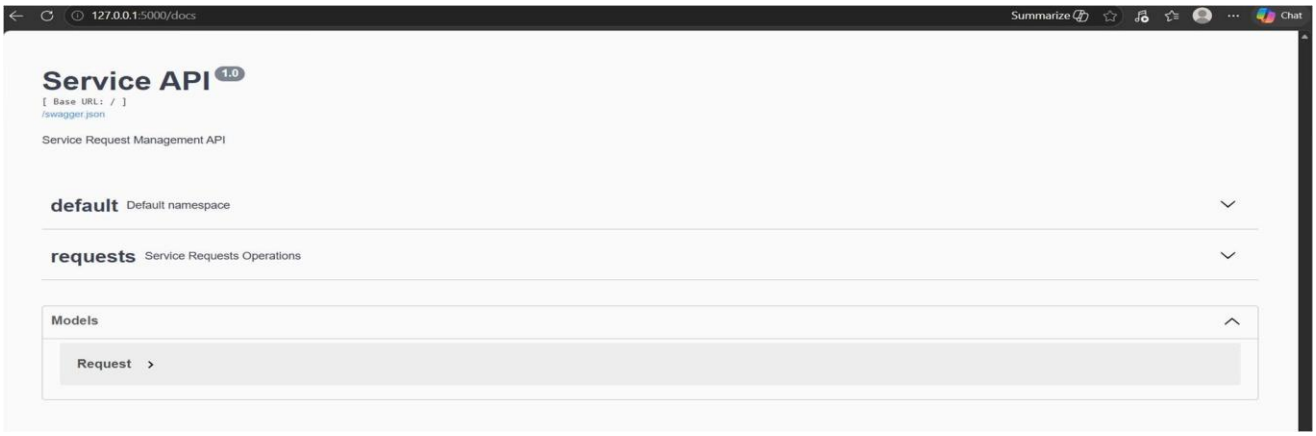
Code:

```
#Add detailed docstrings and inline comments to all Flask API endpoints describing method, inputs, and responses clearly. Optionally integrate Swagger documentation using Flask-RESTX to auto-generate API docs UI.
from flask import Flask, request
from flask_restx import Api, Resource, fields

app = Flask(__name__)

# Swagger UI setup
```

```python
api = Api(app, doc='/docs', title='Service API', description='Service Request Management
API')

# Namespace
ns = api.namespace('requests', description='Service Requests Operations')

# In-memory storage
requests = []

# Model (for Swagger validation + structure)
request_model = api.model('Request', {
    'title': fields.String(required=True, description='Title of the request'),
    'priority': fields.String(required=True, description='Priority level
(High/Medium/Low)')
})

@api.route('/status')
class Status(Resource):
    def get(self):
        """Check API status"""
        return {"status": "API is running"}


@ns.route('/')
class RequestList(Resource):

    def get(self):
        """
        GET /requests
        Retrieve all service requests
        """
        return requests

    @ns.expect(request_model)
    def post(self):
        """
        POST /requests
        Add a new service request
        """
        data = request.get_json()

        new_request = {
            "title": data['title'],
            "priority": data['priority']
        }

        requests.append(new_request)

        return {
            "message": "Request added successfully",
            "request": new_request
        }
```

```python
@ns.route('/<int:index>')
class RequestItem(Resource):

    @ns.expect(request_model)
    def put(self, index):
        """
        PUT /requests/<index>
        Update an existing request
        """
        if index < 0 or index >= len(requests):
            return {"error": "Request not found"}, 404

        data = request.get_json()

        requests[index]["title"] = data.get("title", requests[index]["title"])
        requests[index]["priority"] = data.get("priority", requests[index]["priority"])

        return {
            "message": "Request updated",
            "request": requests[index]
        }

    def delete(self, index):
        """
        DELETE /requests/<index>
        Delete a service request
        """
        if index < 0 or index >= len(requests):
            return {"error": "Request not found"}, 404

        deleted_request = requests.pop(index)

        return {
            "message": "Request deleted",
            "request": deleted_request
        }


if __name__ == '__main__':
    app.run(debug=True)
```

Output:



Explanation:

This Flask application uses Flask-RESTX to implement CRUD operations with automatic Swagger documentation and input validation. It provides an interactive UI where developers can test endpoints and understand request/response formats without reading the code.